

03/7/17

Date : _____
Page No : _____

RELATIONS AND FUNCTIONS

- * Consider, $A = \{1, 2, 3\}$
 Cartesian product on $A = A \times A$
 $A \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
 $n(A) = 3 \therefore n(A \times A) = 3 \times 3 = 9$
- * Relation $\subseteq A \times A$
 Eg: $R = \{(x, y) : x \leq y\}$
 $= \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

Types of Relations:

- i] Reflexive relation : $\forall a \in A, (a, a) \in R$
- ii] Symmetric relation : $\forall (a, b) \in R \Rightarrow (b, a) \in R$
- iii] Transitive relation : $\forall (a, b), (b, c) \in R \Rightarrow (a, c) \in R$
- iv] Equivalence relation : A relation which is call R, S, T.

Eg: *] $A \times A$ on set A

*] ' $\parallel$ ' on set of straight lines

*] '=' on set of $\mathbb{R}$

*] ' $\cong$ ' on set of triangles

*] ' $\simeq$ ' on set of triangles

EXERCISE 1.1

- q] Show that each of the relation R in the set $A = \{x \in \mathbb{Z}, 0 \leq x \leq 12\}$ given by (i) $R = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$
- Sol] Gn: $R = \{(a, b) : |a - b| \text{ is a multiple of } 4\}$
 $= \{(a, b) : |a - b| = 4k\}$
- Let $a, b, c \in A$
- (i) Reflexive property : $\forall a \in A, (a, a) \in R$
 $\forall a \in A, |a - a| = 0 = 4(0) \in A$ is true.
 $\therefore R$ is reflexive $\rightarrow$ ①

(ii) Symmetric property : $\forall (a, b) \in R \Rightarrow (b, a) \in R$

$$\forall (a, b) \in R \Rightarrow |a - b| = 4k$$

$$\Rightarrow |-(a - b)| = 4k$$

$$\Rightarrow |b - a| = 4k \Rightarrow (b, a) \in R$$

$\{(a, a), (a, b), (b, a), (b, b)\} = A \times A$
 $\Rightarrow R$ is symmetric $\rightarrow$ ②

(iii) Transitive property : $\forall (a, b), (b, c) \in R \Rightarrow (a, c) \in R$

$$\forall (a, b), (b, c) \in R \Rightarrow |a - b| = 4k \text{ and } |b - c| = 4l$$

$$\Rightarrow |a - b + b - c| = 4(k + l)$$

$$\{(a, a), (a, b), (b, a), (b, b)\} = A \times A$$

$$\Rightarrow |a - c| = 4m$$

$$\Rightarrow R \text{ is transitive} \rightarrow$$
 ③

①, ②, ③ $\Rightarrow R$ is an equivalence relation

Egs] $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ on $\mathbb{Z}$

$$R = \{(a, b) : a - b = 2k\}$$

T.E. Let $a, b, c \in A$

(i) Reflexive property : $\forall a \in A, (a, a) \in R$

$$\forall a \in A, a - a = 2k \in A \text{ is true}$$

$$\Rightarrow R \text{ is reflexive} \rightarrow$$
 ①

(ii) Symmetric property : $\forall (a, b) \in R \Rightarrow (b, a) \in R$

$$\forall (a, b) \in R \Rightarrow (a - b) = 2k$$

$$\Rightarrow -(a - b) = -2k$$

$$\Rightarrow b - a = -2k \Rightarrow (b, a) \in R$$

$$\therefore R \text{ is symmetric} \rightarrow$$
 ②

(iii) Transitive property : $\forall (a, b), (b, c) \in R \Rightarrow (a, c) \in R$

$$\forall (a, b), (b, c) \in R \Rightarrow (a - b) = 2k \text{ and } (b - c) = 2l$$

$$\Rightarrow a - b + b - c = 2(k + l)$$

$$\Rightarrow a - c = 2m$$

$$\Rightarrow R \text{ is transitive} \rightarrow$$
 ③

①, ②, ③ $\Rightarrow R$ is an equivalence relation

9] ii] $R = \{(a, b) : a = b\}$ on $A = \{x \in \mathbb{Z}, 0 \leq x \leq 12\}$

(i) Reflexive property: $\forall a \in A, (a, a) \in R$

$\forall a \in A, a = a$ is true

Hence, R is reflexive $\rightarrow$ ①

(ii) Symmetric property: $\forall (a, b) \in R \Rightarrow a = b$

$\Rightarrow b = a$ is true and $\therefore (b, a) \in R$

$\therefore R$ is symmetric $\rightarrow$ ②

(iii) Transitive property: $\forall (a, b), (b, c) \in R \Rightarrow (a, c) \in R$

$\forall (a, b), (b, c) \in R \Rightarrow (a, c) \text{ should } \in R$

i.e., $a = b$ and $b = c$ in R

$\Rightarrow a = c$ is true

$\Rightarrow (a, c) \in R$

$\therefore R$ is transitive $\rightarrow$ ③

①, ②, ③ $\Rightarrow R$ is an equivalence relation.

Eg3] Let L be the set of all lines in a plane and R be the relation in L defined as $R = \{(l_1, l_2) : l_1 \perp l_2\}$. Show that R is symmetric but neither reflexive nor transitive.

Sol) Gn: $R = \{(l_1, l_2) : l_1 \perp l_2\}$ on L

Let $l_1, l_2, l_3 \in L$

(i) Reflexive: $\forall a \in L, (a, a) \in R$

$\forall a \in L, l_1 \perp l_1$ is not true

($\because$ a line can't be $\perp$ to itself)

$\therefore R$ is not reflexive

(ii) Symmetric: $\forall (a, b) \in R, (b, a) \in R$

$\forall (l_1, l_2) \in R, \Rightarrow l_1 \perp l_2$

$\Rightarrow l_2 \perp l_1$ is true on L

$\therefore R$ is symmetric

(iii) Transitive : $\forall (a,b), (b,c) \in R \Rightarrow (a,c) \in R$

$\forall (l_1, l_2), (l_2, l_3) \in R \Rightarrow l_1 \parallel l_2$ and $l_2 \parallel l_3$

$\Rightarrow l_1 \parallel l_3$

$\Rightarrow R$ is not transitive

Hence given R is only symmetric, neither reflexive nor transitive.

14] Let L be set of all lines and R be the relation in L defined as $R = \{(L_1, L_2) : L_1 \parallel L_2\}$. Show that R is equivalence relation. Find the set of all lines related to the line $y = 2x + 4$.

Sol] Gn: $R = \{(L_1, L_2) : L_1 \parallel L_2\}$

(i) Reflexive :

$\forall L_1 \in L, L_1 \parallel L_1$ is true

$\therefore R$ is reflexive $\rightarrow$ ①

(ii) Symmetric

$\forall (L_1, L_2) \in R, L_1 \parallel L_2$

$\Rightarrow L_2 \parallel L_1$ is true on L

$\therefore R$ is symmetric $\rightarrow$ ②

(iii) Transitive :

$\forall (L_1, L_2), (L_2, L_3) \in R \Rightarrow L_1 \parallel L_2$ and $L_2 \parallel L_3$

$\Rightarrow L_1 \parallel L_3$

$\Rightarrow R$ is transitive $\rightarrow$ ③

①, ②, ③ $\Rightarrow R$ is an equivalence relation

All the lines related to the line $y = 2x + 4$ and $y = 2x + k$ where k is a real number.

12] Show that R defined in set A of all triangles as $R = \{(T_1, T_2) : T_1 \sim T_2\}$ is equivalence relation. Consider 3 right triangles T_1 with sides 3, 4, 5, T_2 with sides 5, 12, 13 and T_3 with sides 6, 8, 10. Which triangles among T_1, T_2, T_3 are related?

Date : _____
Page No : _____

Sol.] $Gm : R = \{(T_1, T_2) : T_1 \sim T_2\}$

(i) Reflexive : $\forall T_1 \in A, T_1 \sim T_1$ is true
 $\therefore R$ is reflexive. $\rightarrow$ ①

(ii) Symmetric : $\forall (T_1, T_2) \in R, T_1 \sim T_2 \Rightarrow T_2 \sim T_1$ is true
 $\therefore R$ is symmetric. $\rightarrow$ ②

(iii) Transitive : $\forall (T_1, T_2), (T_2, T_3) \in R, T_1 \sim T_2$ and $T_2 \sim T_3 \Rightarrow T_1 \sim T_3$ is true
 $\therefore R$ is transitive. $\rightarrow$ ③

①, ②, ③ $\Rightarrow R$ is an equivalence relation.

Since two Δ 's are similar if corresponding sides are proportional.

$$\therefore \frac{3}{6} = \frac{4}{8} = \frac{5}{10} = \frac{1}{2}$$

$\therefore T_1$ and T_3 are related.

15] Let R be the relation in $A = \{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 2), (2, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}$. Find whether R is R, S, T.

Sol.] i) $\forall a \in A, (a, a) \in R$
 $\therefore R$ is reflexive.

(ii) $(1, 2) \in R$ but $(2, 1) \notin R$
 $\therefore R$ is not symmetric.

(iii) $\forall (a, b), (b, c) \in R, (a, c) \in R$
 $\therefore R$ is transitive.

** 10] Give an eg of Relation which is

- i) S but neither R nor T :
on $A = \{1, 2, 3\}$, $R = \{(2, 3), (3, 2)\}$
- ii) T but neither R nor S :
on $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 3), (1, 3)\}$
- iii) R and S but not T :
on $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$
 $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R$
 $\therefore R$ is not transitive.
- iv) R and T but not S :
on $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$
- v) S and T but not R :
on $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1), (1, 1)\}$

1) Determine whether the following relations are R, S, T :

- i) on $A = \{1, 2, 3, 4, \dots, 14\}$ defined as
 $R = \{(x, y) : 3x - y = 0\}$

Sol) (i) Reflexive property:

$(1, 1) \in A$, $3(1) - 1 \neq 0$
 $\therefore R$ is not reflexive.

(ii) Symmetric:

$(1, 3) \in R \nRightarrow (3, 1) \in R$
 $\therefore R$ is not symmetric.

(iii) Transitive:

$(1, 3), (3, 9) \in R$, but $(1, 9) \notin R$

$\therefore (1, 9)$ does not satisfy $3x - y = 0$

$\therefore R$ is not Transitive.

NOTE: Polynomial functions are never R nor S nor T in general

- ii) R in $\mathbb{N}$ defined by $R = \{(x, y) : y = x + 5, x \leq 4\}$

Sol) Being polynomial, R is neither R nor S nor T .

Date : _____
Page No : _____

iii) Relation R in $A = \{1, 2, 3, 4, 5, 6\}$; $R = \{x, y : y \text{ is divisible by } x\}$

(i) Reflexive :
 x divides x , is true.

(ii) Symmetric :
Eg: 2 divides 4 but 4 does not divide 2
Eg: 2 divides 4 but 4 does not divide 2

(iii) Transitive :
 $x|y, y|z \Rightarrow x|z$ is true on $\mathbb{N}$
Eg: $2|4, 4|8 \Rightarrow 2|8$

iv) On $\mathbb{Z}$, $R = \{x, y : x - y \text{ is an integer}\}$

(i) Reflexive :
 $x - x = 0 \in \mathbb{Z}$ $\therefore R$ is reflexive.

(ii) Symmetric :
 $x - y \in \mathbb{Z} \Rightarrow y - x \in \mathbb{Z} \therefore R$ is symmetric.

(iii) Transitive :
 $x - y \in \mathbb{Z}, y - z \in \mathbb{Z} \Rightarrow x - z \in \mathbb{Z} \therefore R$ is transitive.

kv) Relation R on the set of human beings in a town given by

(a) $R = \{x, y : x \text{ \& } y \text{ work at the same place}\}$

Reflexive $\checkmark$
Symmetric $\checkmark$
Transitive $\checkmark$

(c) $R = \{x, y : x \text{ is exactly } 7 \text{ cm taller than } y\}$

Reflexive $\times$
Symmetric $\times$
Transitive $\times$

(d) $R = \{x, y\} : x \text{ is wife of } y\}$
 Reflexive $\times$
 Symmetric $\times$
 Transitive $\times$

2] Show that R in the set $\mathbb{R}$ of real no.s, defined as $R = \{(a, b) : a \leq b^2\}$ is neither reflexive nor symmetric nor transitive.

Sol] (i) Reflexive:
 $\forall (a, a) \in R, a \leq a^2$ is false for real no.s
 $\therefore R$ is not reflexive.

(ii) Symmetric:
 $\forall (a, b) \in R, a \leq b^2$
 $\nRightarrow b \leq a^2$
 $\therefore R$ is not symmetric.

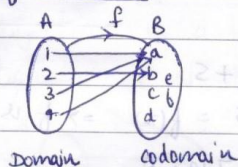
(iii) Transitive:
 $\forall (a, b), (b, c) \in R, a \leq b^2, b \leq c^2$
 $\Rightarrow a \leq c^4$ which is false
 $\therefore R$ is not transitive.

4] Show that R in $\mathbb{R}$ defined as $R = \{(a, b) : a \leq b\}$ is R and T but not S .

Sol] (i) Reflexive:
 $\forall (a, a) \in R, a \leq a$ is true
 $\therefore R$ is reflexive.

(ii) Symmetric:
 $\forall (a, b) \in R, a \leq b$
 $\nRightarrow b \leq a$
 $\therefore R$ is not symmetric.

(iii) Transitive:
 $\forall (a, b), (b, c) \in R, a \leq b, b \leq c$
 $\Rightarrow a \leq c$ is true
 $\therefore R$ is transitive.

Types of Functions:

$$\text{Range} = \{a, b\}$$

$$\text{Range} \in \text{CD}$$

Types: ① Identity $y = f(x) = x$

② Constant: $f(x) = k$

③ Polynomial: $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} \dots + a_n$

$$\text{eg: } y = x^2$$

$$y^2 = x$$

$$y = x^3$$

④ ~~more~~ Modulus: $f(x) = |x| = \begin{cases} x; & x \geq 0 \\ -x; & x < 0 \end{cases}$

⑤ Rational: $\frac{f(x)}{g(x)}$; $g(x) \neq 0$

⑥ Signum: $f(x) = \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$

⑦ Greatest integer: $f(x) = [x]$

Odd and Even Functions:

Odd funⁿ: If $f(-x) = -f(x)$, then $f(x)$ is an odd funⁿ

$$\text{Eg: } f(x) = x^5 - 5x^3 + 7x$$

$$f(-x) = (-x)^5 - 5(-x)^3 + 7(-x) = -x^5 + 5x^3 - 7x$$

$$= -(x^5 - 5x^3 + 7x) = -f(x) \therefore f \text{ is odd}$$

$$f(0) = 2 \sin 30^\circ - \operatorname{cosec} 50^\circ + \tan^3 0^\circ$$

$$f(-0) = 2 \sin (-30^\circ) - \operatorname{cosec} (-50^\circ) + (\tan(-0^\circ))^3$$

$$= -2 \sin 30^\circ + \operatorname{cosec} 50^\circ - \tan^3 0^\circ$$

$$= -f(0)$$

$$\Rightarrow f \text{ is an odd fun}^n.$$

Even funⁿ : $f(-x) = f(x)$, then $f(x)$ is an even funⁿ.

Eg: $f(x) = 4x^4 - 3x^2 + 5$

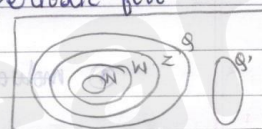
$$f(-x) = 4(-x)^4 - 3(-x)^2 + 5 \\ = 4x^4 - 3x^2 + 5 = f(x) \Rightarrow f \text{ is even}$$

$$f(\theta) = \sin^2 \theta - \cos \theta + 7 \tan^2 \theta$$

$$f(-\theta) = (\sin(-\theta))^2 - \cos(-\theta) + 7(\tan(-\theta))^2 \\ = \sin^2 \theta - \cos \theta + 7 \tan^2 \theta \\ = f(\theta) \therefore f \text{ is even}$$

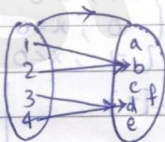
Periodic funⁿ : A funⁿ whose value repeats for every equal no. of period or phase is called periodic funⁿ.

Eg: Trigonometric funⁿ.



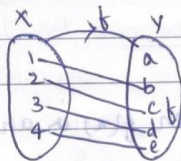
** Types of Functions:

1] Into funⁿ : $\text{Range} \subset \text{CD}$ i.e., free elements present in CD



also Many-one funⁿ

2] Injection or 1-1 function :



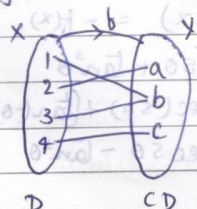
• Presence of one to one correspondence

b/w D and CD

$$f(x_1) = f(x_2)$$

$$\Rightarrow x_1 = x_2$$

3] Surjection or onto funⁿ : $(x) = (y)$



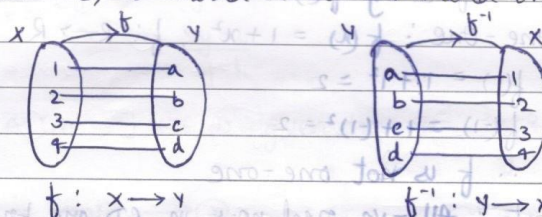
(no free elements in CD)

Range = CD

$$\forall y \in \text{CD}, \exists x \in \text{D}$$

f is onto

- 4] Bijection or 1-1-onto funⁿ:
A bijection is a funⁿ which is both injection and surjection. i.e., it should be both 1-1 and onto.



EXERCISE 1.2

- 7] In each of cases, state whether the funⁿ is 1-1, onto or bijective. Justify.

i) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3 - 4x$

Gn: $f(x) = 3 - 4x$; $f: \mathbb{R} \rightarrow \mathbb{R}$

(a) TPT: f is one-one: $\mathbb{R} = \mathbb{R}$, $\{x\} \rightarrow \mathbb{R} = A \rightarrow B$ [or]

choose $x_1, x_2 \in \mathbb{R}$ of $x_1 \neq x_2$ $\Rightarrow (x_1 - x_2) = (x_1 - x_2)$

Consider $f(x_1) = f(x_2)$

$3 - 4x_1 = 3 - 4x_2$

$-4x_1 = -4x_2$

$x_1 = x_2$

$\therefore f(x_1) = f(x_2) \Rightarrow x_1 = x_2$, $f(x)$ is 1-1 funⁿ $\rightarrow$ ①

(b) TPT: f is onto:

wkt, $y = f(x)$

$y = 3 - 4x$

$x = \frac{3-y}{4}$

① $\leftarrow$ $\mathbb{R} \rightarrow \mathbb{R}$ $x \in \mathbb{R}$ $\Rightarrow (x) \in \mathbb{R}$ $\Rightarrow (x) \in \mathbb{R}$

$\therefore \forall y \in \mathbb{R}, \exists x \in \mathbb{R}$, given by $x = \frac{3-y}{4}$

$\therefore f(x)$ is onto funⁿ $\rightarrow$ ②

① and ② $\Rightarrow$ ~~func~~ exists given by $\boxed{f^{-1}(x) = \frac{3-x}{4}}$

ii] $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 1+x^2$

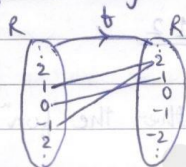
(a) One-One: $f(x) = 1+x^2$; $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(1) = 1+1^2 = 2$$

$$f(-1) = 1+(-1)^2 = 2$$

$\therefore f$ is not one-one

(b) Onto: All -ve real no.s in $\mathbb{C}\mathbb{D}$ are free and do not form images for any element in $\mathbb{D}$.



$\therefore f$ is not onto.

Hence given funcⁿ is neither one-one nor onto.

10] Let $A = \mathbb{R} - \{3\}$, $B = \mathbb{R} - \{1\}$. Consider $f: A \rightarrow B$ defined by $f(x) = \frac{x-2}{x-3}$. Is f one-one and onto. Justify.

(a) One-One:

Choose $(x_1), (x_2)$ from $\mathbb{D}$.

Consider $f(x_1) = f(x_2)$

$$\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3} \quad \text{--- (1)}$$

$$x_1x_2 - 2x_2 - 3x_1 + 6 = x_1x_2 - 2x_1 - 3x_2 + 6$$

$$-2x_2 + 3x_2 = -2x_1 + 3x_1$$

$$x_2 = x_1$$

$\therefore f(x_1) = f(x_2) \Rightarrow x_2 = x_1 \therefore f(x)$ is one-one. $\rightarrow$ ①

(b) onto: Let $y \in \mathbb{C}\mathbb{D}$ then $y \neq 1$.

wkt, $y = f(x)$

$$y = \frac{x-2}{x-3}$$

$xy - 3y = x - 2$
 $xy - x = 3y - 2$
 $x(y - 1) = 3y - 2$
 $x = \frac{3y - 2}{y - 1}$

$\forall y \in \mathbb{C} \setminus \{1\}, \exists x \in \mathbb{C}$ given by $x = \frac{3y - 2}{y - 1}$

$\therefore f$ is onto $\rightarrow (2)$

$(1) \& (2) \Rightarrow f^{-1}(x)$ exists
 given by $f^{-1}(x) = \frac{3x - 2}{x - 1}$

Eg 17] $f(x) = \frac{3x + 4}{5x - 7}$ $f: \mathbb{R} - \left\{\frac{7}{5}\right\} \rightarrow \mathbb{R} - \left\{\frac{3}{5}\right\}$. Show that f is one-one and onto. Write $f^{-1}(x)$.

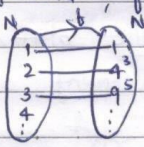
(a) One-One :
 Choose x_1, x_2 from $\mathbb{D}$
 Consider $f(x_1) = f(x_2)$
 $\frac{3x_1 + 4}{5x_1 - 7} = \frac{3x_2 + 4}{5x_2 - 7}$
 $(3x_1 + 4)(5x_2 - 7) = (3x_2 + 4)(5x_1 - 7)$
 $15x_1x_2 - 21x_1 + 20x_2 - 28 = 15x_1x_2 - 21x_2 + 20x_1 - 28$
 $-21x_1 + 20x_2 = -21x_2 + 20x_1$
 $x_1 = x_2$
 $\therefore f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \therefore f(x)$ is one-one $\rightarrow (1)$

(b) onto :
 wkt, $y = f(x)$
 $y = \frac{3x + 4}{5x - 7}$
 $5xy - 7y = 3x + 4$
 $5xy - 3x = 4 + 7y$
 $x(5y - 3) = 7y + 4$
 $x = \frac{7y + 4}{5y - 3}$

$\forall y \in \mathbb{C}D, \exists x \in \mathbb{D}$ given by $x = \frac{7y+4}{3y-5}$
 $\therefore f$ is onto $\rightarrow (2)$
 $(1), (2) \Rightarrow f^{-1}(x)$ exists given by $f^{-1}(x) = \frac{7x+4}{3x-5}$

2] Check injectivity and surjectivity of $f: \mathbb{D} \rightarrow \mathbb{C}D$

i) $f: \mathbb{N} \rightarrow \mathbb{N}; f(x) = x^2$

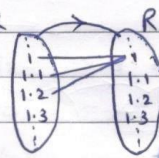

 Injective, Surjective
 f is one-one but not onto
 $\therefore$ Non-perfect squares $3, 5, \dots$ are not images to any element in $\mathbb{D}$

ii) $f: \mathbb{Z} \rightarrow \mathbb{Z}; f(x) = x^2$ $\{x \in \mathbb{Z} \mid x \neq 0\}$ $\therefore f(1) = 1 = (-1)^2$
 iii) $f: \mathbb{R} \rightarrow \mathbb{R}; f(x) = x^2$ $\{x \in \mathbb{R} \mid x \neq 0\}$ $f(1) = 1 = (-1)^2$
 iv) $f: \mathbb{N} \rightarrow \mathbb{N}; f(x) = x^2$ I $\mathbb{N}$ $\mathbb{N}$
 v) $f: \mathbb{Z} \rightarrow \mathbb{Z}; f(x) = x^2$ I $\mathbb{Z}$ $\mathbb{Z}$

3] (i) $f: \mathbb{R} \rightarrow \mathbb{R}; f(x) = [x]$ PT its neither 1-1 nor onto.
 (ii) Write all the points of discontinuity of $[x]$ on $[-3, 3]$.
 Sol] (ii) $[x] = n$ for $x \in [n, n+1)$
 $x \in \{-2, -1, 0, 1, 2\}$

(i) One-One: $[1] = 1$
 $[1.1] = 1$
 $[1.5] = 1$

$\Rightarrow [x]$ is not 1-1

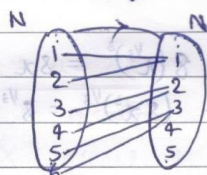
Onto: $\mathbb{R} \rightarrow \mathbb{R}$

 Nos like $1.2, 1.3, \dots$ i.e. all the rationals in $\mathbb{C}D$ do not form images for any element in $\mathbb{D}$.
 $\therefore [x]$ is not 1-1 and also not onto

$x + \mu = (x - \mu)x$
 $\frac{x + \mu}{x - \mu} = x$

9) $f: \mathbb{N} \rightarrow \mathbb{N}; f(n) = \begin{cases} \frac{n+1}{2}; & \text{if } n \text{ is odd} \\ \frac{n}{2}; & \text{if } n \text{ is even} \end{cases} \forall n \in \mathbb{N}$

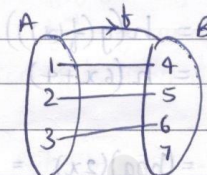
State whether f is bijective. Justify.

Sol.)



From the mapping we observe that $f(n)$ is not 1-1 but is onto. $\therefore f$ is not bijective.

6) $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$, $f = \{(1, 4), (2, 5), (3, 6)\}$ be a funⁿ from A to B . Show that f is 1-1.



Composition of Functions:

* Function inside a funⁿ is called composite funⁿ, like $g \circ f$, $f \circ g$, $f \circ f$, $g \circ g$, $h \circ g \circ f$ (or) $g \circ f \circ g$, $f \circ f \circ f$...
(or) $f \circ g$, $g \circ f$...

Eg: $f(x) = x^2 + 1$, $g(x) = x - 1$. Find $f \circ g$, $g \circ f$, $f \circ f$, $g \circ g$.

(i) $f \circ g(x) = f[g(x)] = f(x-1) = (x-1)^2 + 1$

(ii) $g \circ f(x) = g[f(x)] = g(x^2 + 1) = (x^2 + 1) - 1 = x^2$

(iii) $f \circ f(x) = f[f(x)] = f(x^2 + 1) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$

(iv) $g \circ g(x) = g[g(x)] = g(x-1) = (x-1) - 1 = x - 2$

EXERCISE 1.3

* 3] Find $f \circ g$ and $g \circ f$ if

i) $f(x) = |x|$, $g(x) = (5x-2)$

$$(a) f \circ g(x) = f[g(x)] = f(15x-2) = |15x-2| = |5x-2|$$

$$(b) g \circ f(x) = g[f(x)] = g(1x) = |5x-2|$$

$$ii) f(x) = 8x^3, g(x) = x^{1/3}$$

$$(a) f \circ g(x) = f[g(x)] = f(x^{1/3}) = 8(x^{1/3})^3 = 8x$$

$$(b) g \circ f(x) = g[f(x)] = g(8x^3) = (8x^3)^{1/3} = 8^{1/3}x = \sqrt[3]{8}x = 2x$$

Eg 2b) $f: \mathbb{N} \rightarrow \mathbb{N}, g: \mathbb{N} \rightarrow \mathbb{N}, h: \mathbb{N} \rightarrow \mathbb{R}$ defined as

$f(x) = 2x, g(y) = 3y+4, h(z) = \sin z$. Show that $(h \circ g) \circ f$
 $h \circ (g \circ f)$

$$\text{Sol)} \text{ LHS} = h \circ (g \circ f)(x) = h(g \circ f(x)) = h(g(f(x)))$$

$$= h(g(2x)) = h(3(2x)+4) = h(6x+4)$$

$$= \sin(6x+4) \rightarrow (1)$$

$$\text{RHS} = (h \circ g) \circ f(x) = (h \circ g)(f(x)) = (h \circ g)(2x) = h(g(2x))$$

$$= h(3(2x)+4) = h(6x+4) = \sin(6x+4) \rightarrow (2)$$

$$(1), (2) \rightarrow \text{LHS} = \text{RHS}$$

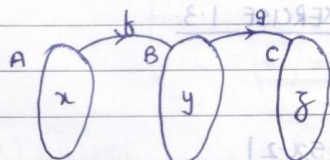
13] If $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = (3-x^3)^{1/3}$, then $f \circ f(x)$ is

$$\text{Sol)} f(f(x)) = f((3-x^3)^{1/3})$$

$$= (3 - ((3-x^3)^{1/3})^3)^{1/3}$$

$$= (3 - (3-x^3))^{1/3} = (x^3)^{1/3} = x$$

Let $f: A \rightarrow B, g: B \rightarrow C$ then the composition of f and g is $g \circ f$ and is defined as the function from A to C and hence $g \circ f: A \rightarrow C$



Let identity element of A be x .

" " " " B be y .

" " " " C be z .

From mapping: $y = f(x)$ and $z = g(y)$
 Consider $z = g(y) = g(f(x)) = g \circ f(x)$
 Eg: Consider $2x \cdot \frac{1}{2} = 1 \Rightarrow 'x'$ is operation
 $'1'$ is identity element under x^n .
 $a * a^{-1} = e$, e is identity element.
 $\downarrow$
 operation
 $\Rightarrow x^{ve}$ inverse of $2 = \frac{1}{2}$
 $\& x^{ve}$ inverse of $\frac{1}{2} = 2 = \frac{1}{\frac{1}{2}}$
 Nly, if $g \circ f(x) = x$ and
 $f \circ g(y) = y$
 Then, $f^{-1} = g$ and $g^{-1} = f$
 Hence f is invertible

Eg 17] Show that if $f: R - \{\frac{1}{5}\} \rightarrow R - \{\frac{3}{5}\}$ is defined by
 $f(x) = \frac{3x+4}{5x-7}$. Show that f is invertible. Write $f^{-1}(x)$.

Sol] Gn: $f: R - \{\frac{1}{5}\} \rightarrow R - \{\frac{3}{5}\}$
 Given $y = f(x) = \frac{3x+4}{5x-7}$
 $\frac{y}{1} = \frac{3x+4}{5x-7}$
 $5xy - 7y = 3x + 4$
 $5xy - 3x = 7y + 4$
 $x(5y - 3) = 7y + 4$
 $x = \frac{7y+4}{5y-3} = g(y)$
 Consider $g \circ f(x) = g[f(x)] = g\left(\frac{3x+4}{5x-7}\right)$
 $= \frac{7\left(\frac{3x+4}{5x-7}\right) + 4}{5\left(\frac{3x+4}{5x-7}\right) - 3}$

$$= \frac{21x + 28 + 20x - 28}{15x + 20 + 15x + 21} = \frac{41x}{41} = x = I_{R-\frac{1}{5}}$$

Consider, $f \circ g(y) = f(g(y)) = f\left(\frac{7y+4}{5y-3}\right)$

$$= \frac{3\left(\frac{7y+4}{5y-3}\right) + 4}{5\left(\frac{7y+4}{5y-3}\right) - 7} = \frac{21y + 12 + 20y - 12}{35y + 20 - 35y + 21} = \frac{41y}{41} = y = I_{R-\frac{1}{5}}$$

$\Rightarrow f$ is invertible

$$\therefore f^{-1}(x) = \frac{7x+4}{5x-3}$$

4] If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$, ST $f \circ f(x) = x$ and f is invertible. Write $f^{-1}(x)$.

Sol] Gn: $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$

$$\text{LHS} = f \circ f(x) = f(f(x)) = f\left(\frac{4x+3}{6x-4}\right) = \frac{4\left(\frac{4x+3}{6x-4}\right) + 3}{6\left(\frac{4x+3}{6x-4}\right) - 4}$$

$$= \frac{16x + 12 + 18x - 12}{24x + 18 - 24x + 16} = \frac{34x}{34} = x = \text{RHS}$$

$$\text{wkt, } y = f(x) = \frac{4x+3}{6x-4} \quad \begin{matrix} x+y = x \\ x-y = x \end{matrix}$$

$$6xy - 4y = 4x + 3$$

$$6xy - 4x = 4y + 3$$

$$x(6y-4) = 4y+3$$

$$x = \frac{4y+3}{6y-4} = g(y)$$

$$x = \frac{4y+3}{6y-4}$$

Date : _____
Page No : _____

Consider $g \circ f(x) = g[f(x)] = g\left(\frac{4x+3}{6x-4}\right)$

$$= \frac{4\left(\frac{4x+3}{6x-4}\right) + 3}{6\left(\frac{4x+3}{6x-4}\right) - 4} = \frac{16x + 12 + 18x - 12}{24x + 18 - 24x + 16}$$

$$= \frac{34x}{34} = x$$

Consider $f \circ g(y) = f[g(y)] = f\left(\frac{4y+3}{6y-4}\right) = \frac{4\left(\frac{4y+3}{6y-4}\right) + 3}{6\left(\frac{4y+3}{6y-4}\right) - 4}$

$$= \frac{16y + 12 + 18y - 12}{24y + 18 - 24y + 16} = \frac{34y}{34} = y$$

$\Rightarrow f$ is invertible and $f^{-1}(x) = \frac{4x+3}{6x-4}$

We infer that

9) Consider $f: \mathbb{R} \rightarrow [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$ s.t. f is invertible. Write $f^{-1}(x)$.

Sol) $y = f(x) = 9x^2 + 6x - 5$

$$y = (3x+1)^2 - 6 \quad \text{Complete square method}$$

$$\frac{y+6}{3} = (3x+1)^2 - 1$$

$$\frac{\sqrt{y+6}-1}{3} = x = g(y)$$

Consider, $g \circ f(x) = g[f(x)] = g[(3x+1)^2 - 6]$

$$= \frac{\sqrt{(3x+1)^2 - 6} - 1}{3} = \frac{3x+1-1}{3} = x = \mathbb{I}_{\mathbb{R}}$$

Consider, $f \circ g(y) = f(g(y)) = f\left(\frac{\sqrt{y+6}-1}{3}\right)$

$$= \left(3\left(\frac{\sqrt{y+6}-1}{3}\right) + 1\right)^2 - 6$$

$$= (\sqrt{y+6}-1+1)^2 - 6$$

$$= y+6-6$$

$$= y = I_{(-5, \infty)}$$

$\Rightarrow f$ is invertible and $f^{-1}(x) = \frac{\sqrt{x+6}-1}{3}$

Eg 25] Let $f: \mathbb{N} \rightarrow \mathbb{R}$ defined by $f(x) = 4x^2 + 12x + 15$ s.t. $f: \mathbb{N} \rightarrow S$ where S is range of f is invertible. Write $f^{-1}(x)$.

Sol] Given: $f: \mathbb{N} \rightarrow S$, $f(x) = 4x^2 + 12x + 15$

wkt, $y = f(x) = 4x^2 + 12x + 15$

$$y = (2x)^2 + 2 \cdot 2x \cdot 3 + 3^2 - 3^2 + 15$$

$$y = (2x+3)^2 - 6 = f(x)$$

$$\frac{\sqrt{y-6}-3}{2} = x = g(y)$$

Consider $g \circ f(x) = g(f(x)) = g((2x+3)^2 - 6)$

$$= \frac{\sqrt{(2x+3)^2 - 6} - 3}{2}$$

$$= \frac{2x+3-3}{2} = \frac{2x}{2} = x = I_{\mathbb{N}}$$

Consider, $f \circ g(y) = f(g(y)) = f\left(\frac{\sqrt{y-6}-3}{2}\right)$

$$I = x = \left(2\left(\frac{\sqrt{y-6}-3}{2}\right) + 3\right)^2 - 6$$

$$= y-6+6 = y = I_S$$

$\Rightarrow f$ is invertible and $f^{-1}(x) = \frac{\sqrt{x-6}-3}{2}$

Eg 23] Let $f: \mathbb{N} \rightarrow \mathbb{Y}$, $f(x) = 4x + 3$, ST f is invertible and find f^{-1} .

Sol] wkt $y = f(x) = 4x + 3$

$$x = \frac{y-3}{4} = g(y)$$

$$\begin{aligned} \text{Consider, } g \circ f(x) &= g(f(x)) = g(4x+3) \\ &= \frac{4x+3-3}{4} = \frac{4x}{4} = x = I_{\mathbb{N}} \end{aligned}$$

$$f \circ g(y) = f(g(y)) = f\left(\frac{y-3}{4}\right) = 4\left(\frac{y-3}{4}\right) + 3 = y - 3 + 3 = y = I_{\mathbb{Y}}$$

$\Rightarrow f$ is invertible, $f^{-1}(x) = \frac{x-3}{4}$

6] f is a funⁿ from $[-1, 1]$ to $\mathbb{R}$ given by $f(x) = \frac{x}{x+2}$.
 ST f is 1-1. Find inverse of f .

Sol] Gn: $f: [-1, 1] \rightarrow \mathbb{R}$

(a) One - One: Choose x_1, x_2 belonging to $\mathbb{D}$

Consider, $f(x_1) = f(x_2)$

$$\Rightarrow \frac{x_1}{x_1+2} = \frac{x_2}{x_2+2}$$

$$\begin{aligned} \Rightarrow x_1(x_2+2) &= x_2(x_1+2) \\ \Rightarrow x_1x_2 + 2x_1 &= x_1x_2 + 2x_2 \\ 2x_1 &= 2x_2 \\ \Rightarrow x_1 &= x_2 \end{aligned}$$

$\therefore f(x_1) = f(x_2) \Rightarrow x_1 = x_2$, $f(x)$ is one - one.

(b) Inverse (onto):

wkt, $y = f(x)$

$$y = \frac{x}{x+2}$$

$$xy + 2y = x$$

$$xy - x = -2y \quad 2y = (x - xy)$$

$$2y = x(1-y)$$

$$x = \frac{2y}{1-y} \quad \therefore f^{-1}(x) = \frac{2x}{1-x}$$

14] Let $f: \mathbb{R} \rightarrow \mathbb{R} \setminus \{-\frac{4}{3}\}$, $f(x) = \frac{4x}{3x+4}$. Inverse of f is given

by (a) $g(y) = \frac{3y}{3-4y}$ (b) $g(y) = \frac{4y}{4-3y}$ (c) $g(y) = \frac{4y}{3-4y}$

(d) $g(y) = \frac{3y}{4-3y}$

Sol] wkt, $y = f(x)$
 $y = \frac{4x}{3x+4}$

$y(3x+4) = 4x$
 $3xy + 4y = 4x$

$4y = (4x - 3xy)$

$4y = x(4 - 3y)$

$x = \frac{4y}{4-3y}$ (b)

8] Consider $f: \mathbb{R}_+ \rightarrow [4, \infty)$ given by $f(x) = x^2 + 4$. ST f is invertible and write $f^{-1}(x)$.

Sol] wkt, $y = f(x)$

$y = x^2 + 4$

$x = \sqrt{y-4} = g(y)$

Consider, $g \circ f(x) = g(f(x)) = g(x^2 + 4) = \sqrt{(x^2 + 4) - 4} = x = I$

$f \circ g(y) = f(g(y)) = f(\sqrt{y-4})$
 $= (\sqrt{y-4})^2 + 4$

$= y - 4 + 4 = y = I_{[4, \infty)}$

$\Rightarrow f$ is invertible, $f^{-1}(x) = \sqrt{x-4}$

Binary Operations:

Eg: $N = \{1, 2, 3, 4, \dots\}$

Let $2, 4 \in N$

$\ast \rightarrow +^m : 2 + 4 = 6 \in N$

$\frac{x \cdot y}{x - y} = (x)^{-1} \cdot y$ $\frac{p \cdot q}{p - q} = x$

Date : _____
Page No : _____

$\Rightarrow '+'$ is b.o on $\mathbb{N}$

$* \rightarrow -^m : 2-4 = -2 \notin \mathbb{N} \therefore -^m$ is not b.o on $\mathbb{N}$

$* \rightarrow -^m$ on $\mathbb{Z} : \forall a, b \in \mathbb{Z}, a-b \in \mathbb{Z}$

$\therefore -^m$ is b.o on $\mathbb{Z}$

A binary operation $*$ on a set A is a function $*$: $A \times A \rightarrow A$ denoted by $a * b$.

EXERCISE 1.4

i) Determine whether or not $*$ given below is b.o on not. Justify

i) On $\mathbb{Z}^+$; $a * b = a - b$

Sol] $\forall a, b \in \mathbb{Z}^+, a * b = a - b \notin \mathbb{Z}^+$

$\therefore *$ is not b.o on $\mathbb{Z}^+ \because 2-4 = -2 \notin \mathbb{Z}^+$

ii) On $\mathbb{Z}^+$; $a * b = ab$

Sol] $\forall a, b \in \mathbb{Z}^+, a * b = ab \in \mathbb{Z}^+ = (1 \times 1) \times 1 = 1$

$\therefore *$ is b.o on $\mathbb{Z}^+ \because (1 \times 1) \times 1 = 1$

iii) On $\mathbb{R}$; $a * b = ab^2$

Sol] $\forall a, b \in \mathbb{R}, a * b = ab^2 \in \mathbb{R}$

$\therefore *$ is b.o on $\mathbb{R}$

iv) On $\mathbb{Z}^+$; $a * b = |a - b|$

Sol] $\forall a, b \in \mathbb{Z}^+, a * b = |a - b| \in \mathbb{R}$

$\therefore *$ is b.o on $\mathbb{Z}^+$

v) On $\mathbb{Z}^+$; $a * b = a(1 + b)$

Sol] $\forall a, b \in \mathbb{Z}^+, a * b = a(1 + b) \in \mathbb{Z}^+ = 1 \times (1 + 1) = 2$

$\therefore *$ is b.o on $\mathbb{Z}^+$

Properties on $\mathcal{BO}$:

(a) Commutative $\div \forall a, b \in A, a * b = b * a$

(b) Associative $\div \forall a, b, c \in A, a * (b * c) = (a * b) * c$

(c) Existence of Identity $\div \forall a \in A, a * e = e * a = a$

[To find id, use $a * e = a$]

(d) Existence of Inverse (a^{-1}) $\div \forall a, a^{-1} \in A, a * a^{-1} = a^{-1} * a = e$

[To find a^{-1} , use $a * a^{-1} = e$]

2] For each operation given below, determine whether $*$ is

(a) $\mathcal{BO}$ (b) commutative (c) associative.

i] On $\mathbb{Z}$, $a * b = a - b$

(a) $\forall a, b \in \mathbb{Z}, a * b = a - b \in \mathbb{Z}$

(b) Commutative law: $\forall a, b \in A, a * b = b * a$

LHS = $a * b = a - b$ } LHS $\neq$ RHS

RHS = $b * a = b - a$ } $\therefore *$ is not commutative

(c) Associative: $\forall a, b, c \in A, a * (b * c) = (a * b) * c$

LHS = $a * (b * c) = a * (b - c) = a - b + c$ } LHS $\neq$ RHS

RHS = $(a * b) * c = (a - b) * c = a - b - c$ } $\therefore *$ is not associative

ii] On $\mathbb{Q}$, $a * b = ab + 1$

(a) $\forall a, b \in \mathbb{Q}, a * b = ab + 1 \in \mathbb{Q}$

$\therefore *$ is $\mathcal{BO}$ on $\mathbb{Q}$

(b) LHS = $a * b = ab + 1$ } LHS = RHS

RHS = $b * a = ba + 1 = ab + 1$ } $\therefore *$ is commutative.

(c) LHS = $a * (b * c) = a * (bc + 1) = abc + a + 1$ } LHS $\neq$ RHS

RHS = $(a * b) * c = (ab + 1) * c = abc + c + 1$ } $\therefore *$ is not associative

iii] On $\mathbb{Q}$, $a * b = \frac{ab}{2}$

(d) Existence of Id : $\forall a, e \in A, a * e = a$ and $e * a = a$ $\forall (a)$
 Consider, $a * e = a$
 $\frac{ae}{4} = a$
 $e = 4$

(e) Existence of inverse : $\forall a, a^{-1} \in A, a * a^{-1} = e$
 Consider, $a * a^{-1} = e$
 $\frac{aa^{-1}}{4} = e = 4$
 $aa^{-1} = 16$
 $a^{-1} = \frac{16}{a}$

v) On $\mathbb{Z}^+$, $a * b = 2^{ab}$
 (a) $\forall a, b \in \mathbb{Z}^+, a * b = 2^{ab} \in \mathbb{Z}^+$
 $\therefore *$ is b.o on $\mathbb{Z}^+$
 (b) LHS = $a * b = 2^{ab}$
 RHS = $b * a = 2^{ba} = 2^{ab}$
 $\therefore *$ is commutative
 (c) LHS = $a * (b * c) = a * 2^{bc} = 2^{a \cdot 2^{bc}}$
 RHS = $(a * b) * c = 2^{ab} * c = 2^{2^{ab} \cdot c}$
 $\therefore *$ is not associative.

vi) On $\mathbb{Z}^+$, $a * b = a^b$
 (a) $\forall a, b \in \mathbb{Z}^+, a * b = a^b \in \mathbb{Z}^+$
 $\therefore *$ is b.o on $\mathbb{Z}^+$
 (b) LHS = $a * b = a^b$
 RHS = $b * a = b^a$
 $\therefore *$ is not commutative
 (c) LHS = $a * (b * c) = a * (b^c) = a^{b^c}$
 RHS = $(a * b) * c = (a^b) * c = (a^b)^c$
 $\therefore *$ is not associative

vii) On $\mathbb{R}$, $a * b = \frac{a}{b+1}$

Date: _____
Page No: _____

(a) $\forall a, b \in R, a * b = \frac{a}{b+1} \in R$
 $\therefore *$ is b.o on R

(b) $LHS = a * b = \frac{a}{b+1}$
 $RHS = b * a = \frac{b}{a+1}$
 $LHS \neq RHS$
 $\therefore *$ is not commutative

(c) $LHS = a * (b * c) = a * \left(\frac{b}{c+1}\right) = \frac{a}{\frac{b}{c+1} + 1}$
 $RHS = (a * b) * c = \left(\frac{a}{b+1}\right) * c = \frac{\frac{a}{b+1}}{\frac{a}{b+1} + 1} = \frac{a}{(b+1)(\frac{a}{b+1} + 1)}$
 $LHS \neq RHS$
 $\therefore *$ is not associative.

3] Consider the b.o ' $\wedge$ ' on the set $\{1, 2, 3, 4, 5\}$ defined by $a \wedge b = \min\{a, b\}$. Write the operation table.

$\wedge$	1	2	3	4	5
1	1	1	1	1	1
2	1	2	2	2	2
3	1	2	3	3	3
4	1	2	3	4	4
5	1	2	3	4	5

4] Consider b.o $*$ on set $\{1, 2, 3, 4, 5\}$ given by

$*$	1	2	3	4	5
1	1	1	1	1	1
2	1	2	1	2	1
3	1	1	3	1	3
4	1	2	1	4	1
5	1	1	1	1	5

(i) Compute $(2 * 3) * 4$ and $2 * (3 * 4)$
 $= 1 * 4$ and $2 * 1$
 $= 1$ and 1
 $\therefore (2 * 3) * 4 = 2 * (3 * 4)$
 $*$ is associative.

(ii) Is $*$ commutative? $a \times b = b \times a$ (a)

Yes, since the table is symmetric about pd.

(iii) Compute $(2 \times 3) \times (4 \times 5)$ (b)

$$\equiv 1 \times 1$$

$$\equiv 1$$

5] Let $*$ be the b.o on $\{1, 2, 3, 4, 5\}$ defined by $a * b = \text{HCF of } a \text{ and } b$. Construct the table.

$*$	1	2	3	4	5
1	1	1	1	1	1
2	1	2	1	2	1
3	1	1	3	1	1
4	1	2	1	4	1
5	1	1	1	1	5

6] Let $*$ be b.o on $\mathbb{N}$ given by $a * b = \text{LCM of } a \text{ and } b$.

Find (i) $5 * 7$

$$= \text{LCM of } 5 \text{ \& } 7 = 35$$

(ii) $20 * 16$

$$= \text{LCM of } 20 \text{ \& } 16$$

$$= 80$$

(iii) Is $*$ commutative?

$$\text{LCM of } a \text{ \& } b = \text{LCM of } b \text{ \& } a$$

$\therefore *$ is commutative

(iv) Is $*$ associative?

$$\text{LHS} = a * (b * c)$$

$$= a * (\text{LCM of } b \text{ \& } c)$$

$$= \text{LCM of } a, b, c \quad (i)$$

$$= \text{LCM of } a, b * c$$

$$= \text{LCM of } (a * b) * c$$

$$= \text{RHS} \quad (ii)$$

$\therefore *$ is associative

Date : _____
Page No : _____

7] Is $*$ defined on the set $\{1, 2, 3, 4, 5\}$ by $a * b = \text{LCM of } a \& b$, a b.o. Justify.

Sol] No, $*$ is not b.o on given set.
 $\therefore \text{LCM of } 3, 4 = 12 \notin \text{set}$

9] Let $*$ be b.o on $\mathbb{Q}$. Find which of the b.o are commutative, associative?

i) $a * b = a - b$

(a) Commutative:
 $\text{LHS} = a * b = a - b \neq b - a = b * a = \text{RHS}$
 $\therefore \text{LHS} \neq \text{RHS}, \therefore *$ is not commutative.

(b) Associative: $*$ is not associative.

ii) $a * b = a^2 + b^2$

(a) Commutative: $\forall a, b \in \mathbb{Q}, a * b = b * a$
 $\text{LHS} = a * b = a^2 + b^2 = b^2 + a^2 = b * a = \text{RHS}$
 $\therefore *$ is commutative.

(b) Associative: $\forall a, b, c \in \mathbb{Q}, (a * b) * c = a * (b * c)$
 $\text{LHS} = (a * b) * c = (a^2 + b^2) * c = (a^2 + b^2)^2 + c^2$
 $\text{RHS} = a * (b * c) = a * (b^2 + c^2) = a^2 + (b^2 + c^2)^2$
 $\text{LHS} \neq \text{RHS}, \therefore *$ is not associative.

iii) $a * b = a + ab$

(a) Commutative: $(b + a, c + a) = (b, c) * (d, a) = 2H1$
 $\text{LHS} = a * b = a + ab$
 $\text{RHS} = b * a = b + ba$
 $\therefore \text{LHS} \neq \text{RHS}, \therefore *$ is not commutative.

(b) Associative: $(b + c, a + c) = (b, c) * (d, a) = 2H1$
 $\text{LHS} = (a * (b * c)) = a * (b + bc) = (a) + a(b + bc)$
 $\text{RHS} = ((a * b) * c) = (a + ab) * c = a + ab + (a + ab)c$
 $\therefore \text{LHS} \neq \text{RHS}, \therefore *$ is not associative.

iv) $a * b = (a-b)^2$

(a) Commutative:

$$\left. \begin{aligned} \text{LHS} &= a * b = (a-b)^2 \\ \text{RHS} &= b * a = (b-a)^2 \end{aligned} \right\} \because \text{LHS} \neq \text{RHS} \therefore * \text{ is not commutative}$$

(b) Associative:

$$\left. \begin{aligned} \text{LHS} &= a * (b * c) = a * (b-c)^2 = [a - (b-c)^2]^2 \\ \text{RHS} &= (a * b) * c = (a-b)^2 * c = [(a-b)^2 - c]^2 \end{aligned} \right\} \because \text{LHS} \neq \text{RHS} \therefore * \text{ is not associative}$$

v) $a * b = ab^2$

(a) Commutative:

$$\left. \begin{aligned} \text{LHS} &= a * b = ab^2 \\ \text{RHS} &= b * a = ba^2 \end{aligned} \right\} \because \text{LHS} \neq \text{RHS} \therefore * \text{ is not commutative}$$

(b) Associative:

$$\left. \begin{aligned} \text{LHS} &= a * (b * c) = a * (bc^2) = a(bc^2)^2 \\ \text{RHS} &= (a * b) * c = (ab^2) * c = ab^2c^2 \end{aligned} \right\} \because \text{LHS} \neq \text{RHS} \therefore * \text{ is not associative}$$

ii) Let $A = \mathbb{N} \times \mathbb{N}$ and $*$ be b.o defined by $(a, b) * (c, d) = (a+c, b+d)$.
 (ST) $*$ is commutative and associative. Find e if any.

Sol) Given: $(a, b) * (c, d) = (a+c, b+d)$ on A

(a) Commutative:

Let $(a, b), (c, d), (e, f)$ be 3 pairs $\in A$

$$\begin{aligned} \text{LHS} &= (a, b) * (c, d) = (a+c, b+d) \\ \text{RHS} &= (c, d) * (a, b) = (c+a, d+b) \\ &= (a+c, b+d) \\ \text{LHS} &= \text{RHS}, \therefore * \text{ is commutative} \end{aligned}$$

(b) Associative:

$$\begin{aligned} \text{LHS} &= (a, b) * [(c, d) * (e, f)] = (a, b) * (c+e, d+f) \\ &= (a+c+e, b+d+f) \\ \text{RHS} &= [(a, b) * (c, d)] * (e, f) = (a+c, b+d) * (e, f) \\ &= (a+c+e, b+d+f) \\ \text{LHS} &= \text{RHS}, \therefore * \text{ is associative} \end{aligned}$$

(e) Existence of ~~inver~~ identity element : $\forall a, e \in A, a * e = a$

$$(a, b) * (e, e) = (a, b)$$

$$(a+e, b+e) = (a, b)$$

$$a+e = a \quad | \quad b+e = b$$

$$e = 0 \quad | \quad e = 0 \notin \mathbb{N}$$

$\therefore$ identity element does not exist for the given operation on the given set.

10] Find which of the operations in Q9 have identity.

i) $a * b = a - b$

$$\forall a, e \in A, a * e = a$$

$$\text{Consider, } a * e = a$$

$$a - e = a$$

$$e = 0$$

ii) $a * b = a^2 + b^2$

$$\text{Consider, } a * e = a$$

$$a^2 + e^2 = a$$

$$e^2 = a - a^2$$

$$e = \sqrt{a - a^2}$$

iii) $a * b = a + ab$

$$\text{Consider, } a * e = a$$

$$a + ae = a$$

$$ae = 0$$

$$e = 0$$

iv) $a * b = (a - b)^2$

$$a * e = a$$

$$(a - e)^2 = a$$

$$a - e = \sqrt{a}$$

$$e = a - \sqrt{a}$$

v) $a * b = ab^2$

$$a * e = a$$

$$ae^2 = a$$

$$e^2 = 1$$

$$e = 1$$